

DELHI, ON, Canada

WMO: 715730

Lat: 42.8635N

Lon: 80.5453W

Elev: 760

StdP: 14.30

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.2	(c) 2.5	(d) -9.7	(e) 3.4	(f) -1.6	(g) -3.6	(h) 4.7	(i) 3.3	(j) 18.9	(k) 29.1	(l) 17.3	(m) 29.5	(n) 3.7	(o) 260	(p) 0.491

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	87.3	71.4	84.7	70.3	82.2	69.1	75.1	82.3	73.4	80.4	71.8	78.6	8.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.0	125.9	78.6	71.1	118.1	77.2	69.5	111.7	75.5	39.1	82.5	37.5	80.4	36.0	79.3	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	14.4	12.9	DB	-11.8	91.0	8.5	3.0	-17.9	93.2	-22.9	94.9	-27.7	96.6	-33.9	98.7
			WB	-12.2	78.1	8.3	2.5	-18.2	79.9	-23.1	81.3	-27.7	82.8	-33.8	84.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.2	22.7	23.9	32.8	44.6	56.6	65.9	70.0	68.2	61.8	50.1	39.5	29.0	
	DBStd	18.80	11.70	10.74	10.83	8.76	8.31	6.38	5.51	5.16	7.00	8.23	8.49	9.47	
	HDD50	3443	848	731	543	206	29	0	0	0	3	103	328	652	
	HDD65	6958	1311	1151	999	612	285	65	17	27	144	466	765	1116	
	CDD50	2424	1	0	9	45	234	476	619	563	356	107	13	1	
	CDD65	464	0	0	0	1	25	91	170	125	47	5	0	0	
	CDH74	4168	0	0	3	17	300	839	1559	1003	419	28	0	0	
	CDH80	1040	0	0	0	0	65	220	448	212	93	2	0	0	
Wind	WSAvg	5.3	6.2	6.0	5.8	5.9	5.3	5.0	4.6	4.3	4.2	5.1	5.7	5.8	
Precipitation	PrecAvg	38.3	2.3	2.2	3.2	3.5	3.1	3.2	3.0	3.6	3.5	3.0	3.8	3.5	
	PrecMax	47.1	4.7	4.9	7.6	6.0	5.6	6.6	7.3	10.5	11.5	6.4	7.8	5.7	
	PrecMin	24.8	0.7	0.6	0.5	1.5	1.0	0.6	1.3	0.4	0.8	0.5	1.2	1.5	
	PrecStd	5.1	1.0	1.1	1.6	1.1	1.4	1.3	1.4	2.2	2.2	1.4	1.6	1.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.9	54.7	69.0	76.5	85.5	88.7	91.3	89.0	87.0	77.7	64.9	57.1	
		MCWB	53.0	49.7	59.5	62.0	67.5	70.5	73.8	73.6	71.2	66.1	56.6	53.6	
	2%	DB	48.7	48.2	59.3	70.8	81.5	85.7	87.8	85.0	82.6	72.1	60.5	50.4	
		MCWB	46.1	45.2	51.2	57.0	65.3	69.6	72.2	71.7	70.4	63.2	54.9	47.5	
	5%	DB	42.2	42.4	53.7	65.5	77.3	82.4	84.9	82.1	78.9	68.2	57.1	45.7	
		MCWB	39.7	39.0	48.5	54.3	63.9	69.1	70.5	69.9	67.9	60.8	52.4	43.4	
	10%	DB	37.5	37.4	48.7	60.4	72.4	78.8	82.1	79.5	75.2	64.1	53.4	41.2	
		MCWB	35.5	34.7	43.7	51.2	60.2	65.8	69.1	68.8	66.0	58.4	49.2	39.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.0	51.6	60.0	65.7	71.0	74.9	78.2	77.3	75.1	69.8	59.6	53.3
			MCDB	55.8	53.5	69.1	72.1	80.2	84.7	86.0	83.9	82.5	75.2	62.5	55.9
2%		WB	46.5	45.2	53.6	59.8	68.1	72.5	75.5	74.7	72.6	65.6	56.1	48.2	
		MCDB	47.9	48.4	57.7	66.6	78.3	81.5	83.3	80.6	78.8	69.9	60.0	49.7	
5%		WB	40.0	39.2	48.7	56.3	65.7	70.5	73.5	72.9	70.1	61.9	52.6	43.5	
		MCDB	41.8	42.2	52.9	63.5	74.1	78.7	80.5	79.0	75.8	66.3	56.0	45.6	
10%		WB	35.8	35.1	43.8	52.3	62.9	68.6	71.7	71.1	67.7	59.0	49.5	39.3	
		MCDB	37.2	37.3	48.2	59.1	69.8	75.9	78.9	76.8	73.4	63.2	53.1	41.0	
Mean Daily Temperature Range		5% DB	MDBR	13.8	15.1	16.1	19.8	21.3	20.2	21.0	20.3	20.6	17.1	14.5	11.9
			MCDBR	16.6	20.1	24.5	29.1	26.6	23.9	23.4	22.8	25.1	22.6	20.8	16.3
	MCWBR		15.5	16.9	17.6	18.0	13.5	11.3	10.9	11.8	13.4	14.0	15.8	14.6	
	5% WB	MCDBR	15.7	19.8	22.5	23.7	23.4	20.8	19.5	18.5	20.1	19.3	18.8	16.1	
		MCWBR	15.2	17.3	17.6	17.3	13.7	11.7	10.3	10.8	12.1	13.9	15.3	14.8	
Clear-Sky Solar Irradiance	taub	0.294	0.342	0.334	0.372	0.397	0.430	0.422	0.407	0.379	0.358	0.313	0.291		
	taud	2.396	2.202	2.366	2.320	2.298	2.235	2.279	2.307	2.369	2.410	2.504	2.474		
	Ebn at Noon	268	267	288	284	278	268	269	270	269	260	260	260		
	Edh at Noon	25	36	34	39	41	43	41	39	34	29	23	21		
All-Sky Solar Radiation	RadAvg	470	676	1082	1432	1761	1959	1945	1703	1345	821	506	314		
	RadStd	41	90	128	153	166	128	132	87	124	77	72	34		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	-1.44	-2.07	N/A	N/A	N/A	N/A
(47)	Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page